Class Notes

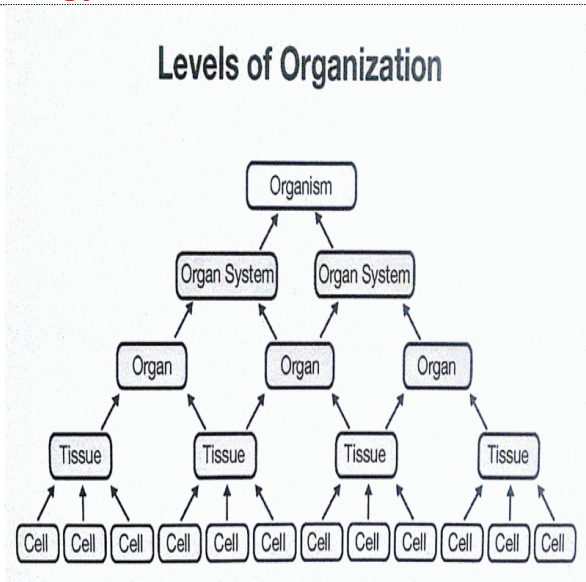
Class:XI	Topic: CHAPTER 8 CELL: THE UNIT OF LIFE…PART 1
Subject: BIOLOGY	

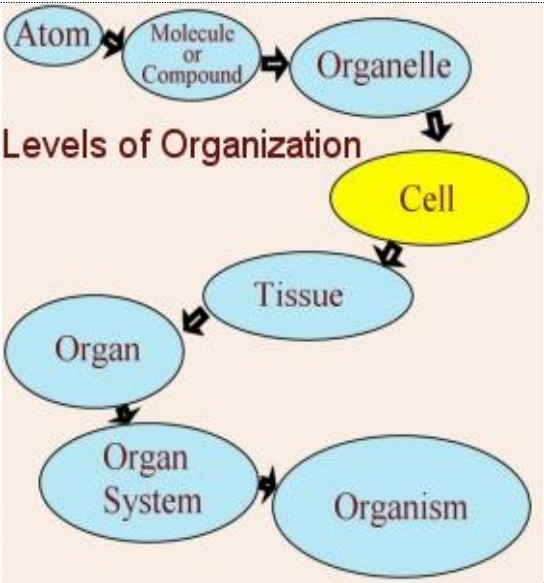
- **CELL DEFINITION**
- **WHAT IS A CELL?(CHARACTERISTICS OF CELLS)**
- **DISCOVERY**
- **CELL THEORY**
- **FUNCTIONS OF A CELL**
- **PROKARYOTIC CELLS**
- **EUKARYOTIC CELLS**

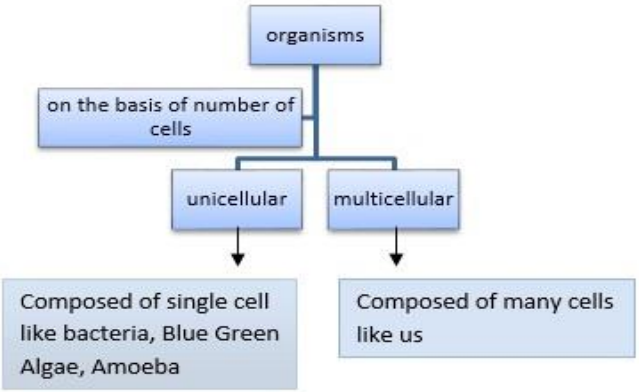
Cell Definition:

"A cell is defined as the smallest, basic unit of life that is responsible for all of life's processes."

The study of cells from its basic structure to the functions of every cell organelle is called Cell Biology.







All organisms are made up of cells.
They may be made up of a single cell (unicellular), or many cells (multicellular).

What is a Cell?

Cells are the basic, fundamental unit of life.
So, if we were to break apart an organism to the cellular level, the smallest independent component that we would find would be the cell.
Cells are the structural, functional, and biological units of all living beings.
A cell can replicate itself independently. **Hence, they are known as the building blocks of life**.

Each cell contains a fluid called the cytoplasm, which is enclosed by a membrane. Also present in the cytoplasm are several biomolecules like proteins, nucleic acids and lipids. Moreover, cellular structures called cell organelles are suspended in the cytoplasm.

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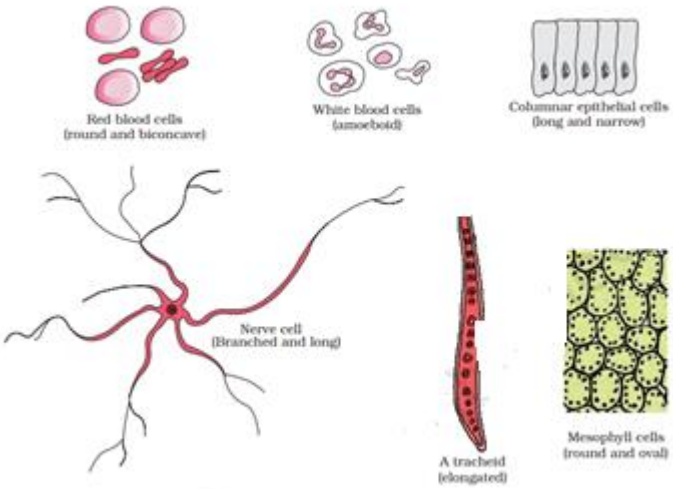
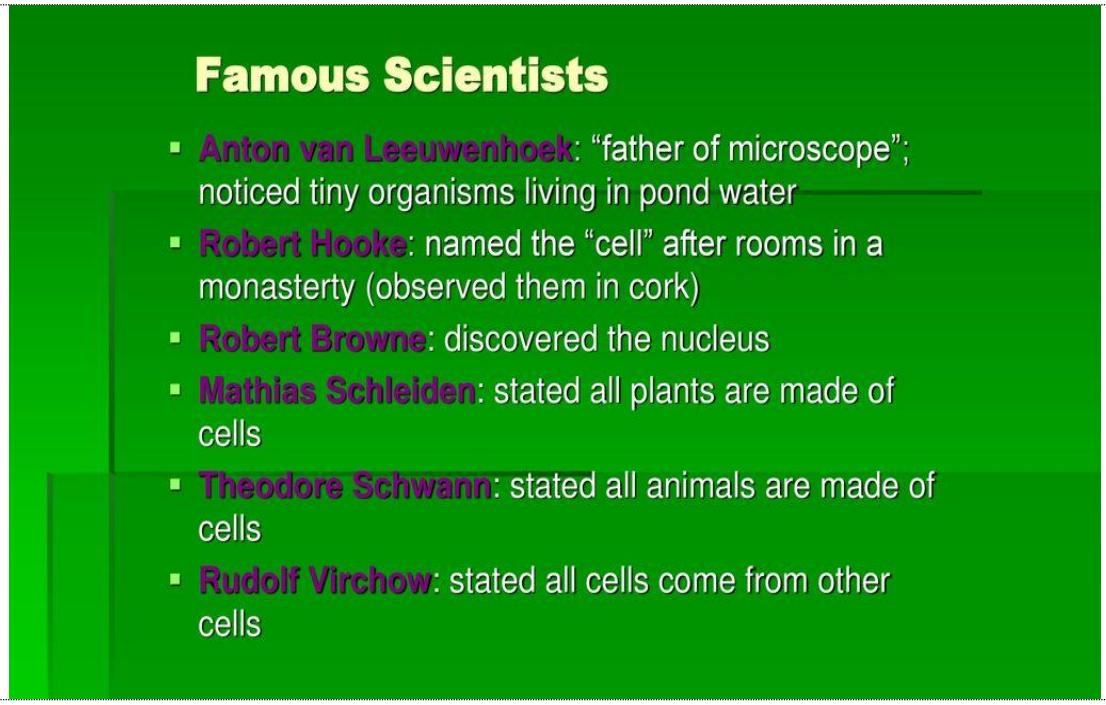


Mycoplasmas are the smallest known cells.
Cells are complex, and their components perform various functions in an organism.
They are of different shapes and sizes, pretty much like bricks of the buildings. Our body is made up of cells of different shapes and sizes.

Cells are the lowest level of organisation in every life form. From organism to organism, the count of cells may vary. Humans have the number of cells compared to that of bacteria.

DISCOVERY OF CELLS:
Discovery of cells is one of the remarkable advancements in the field of science. It helped us know that all the organisms are made up of cells, and these cells help in carrying out various life processes. The structure and functions of cells helped us to understand life in a better way.

Who discovered cells?
Robert Hooke discovered the cell in 1665. Robert Hooke observed a piece of bottle cork under a compound microscope and noticed minuscule structures that reminded him of small rooms. Consequently, he named these “rooms” as cells. However, his compound microscope had limited magnification, and hence, he could not see any details in the structure. Because of this limitation, Hooke concluded that these were non-living entities.
Later Anton Van Leeuwenhoek observed cells under another compound microscope with higher magnification. This time, he had noted that the cells exhibited some form of movement (motility). As a result, Leeuwenhoek concluded that these microscopic entities were “alive.” Eventually, after a host of other observations, these entities were named as animalcules.
In 1883, Robert Brown, a Scottish botanist, provided the very first insights into the cell structure. He was able to describe the nucleus present in the cells of orchids.



Characteristics of Cells:

FOLLOWING ARE THE VARIOUS ESSENTIAL CHARACTERISTICS OF CELLS:

Cells provide structure and support to the body of an organism.
The cell interior is organised into different individual organelles surrounded by a separate membrane.
The nucleus(major organelle) holds genetic information necessary for reproduction and cell growth.
Every cell has one nucleus and membrane-bound organelles in the cytoplasm.
Mitochondria, a double membrane-bound organelle is mainly responsible for the energy transactions vital for the survival of the cell.
Lysosomes digest unwanted materials in the cell.
Endoplasmic reticulum plays a significant role in the internal organisation of the cell by synthesising selective molecules and processing, directing and sorting them to their appropriate locations.

Cell Theory

Cell Theory was proposed by the German scientists:

- **Theodor Schwann**
- **Matthias Schleiden**
- **Rudolf Virchow.**

The cell theory states that:

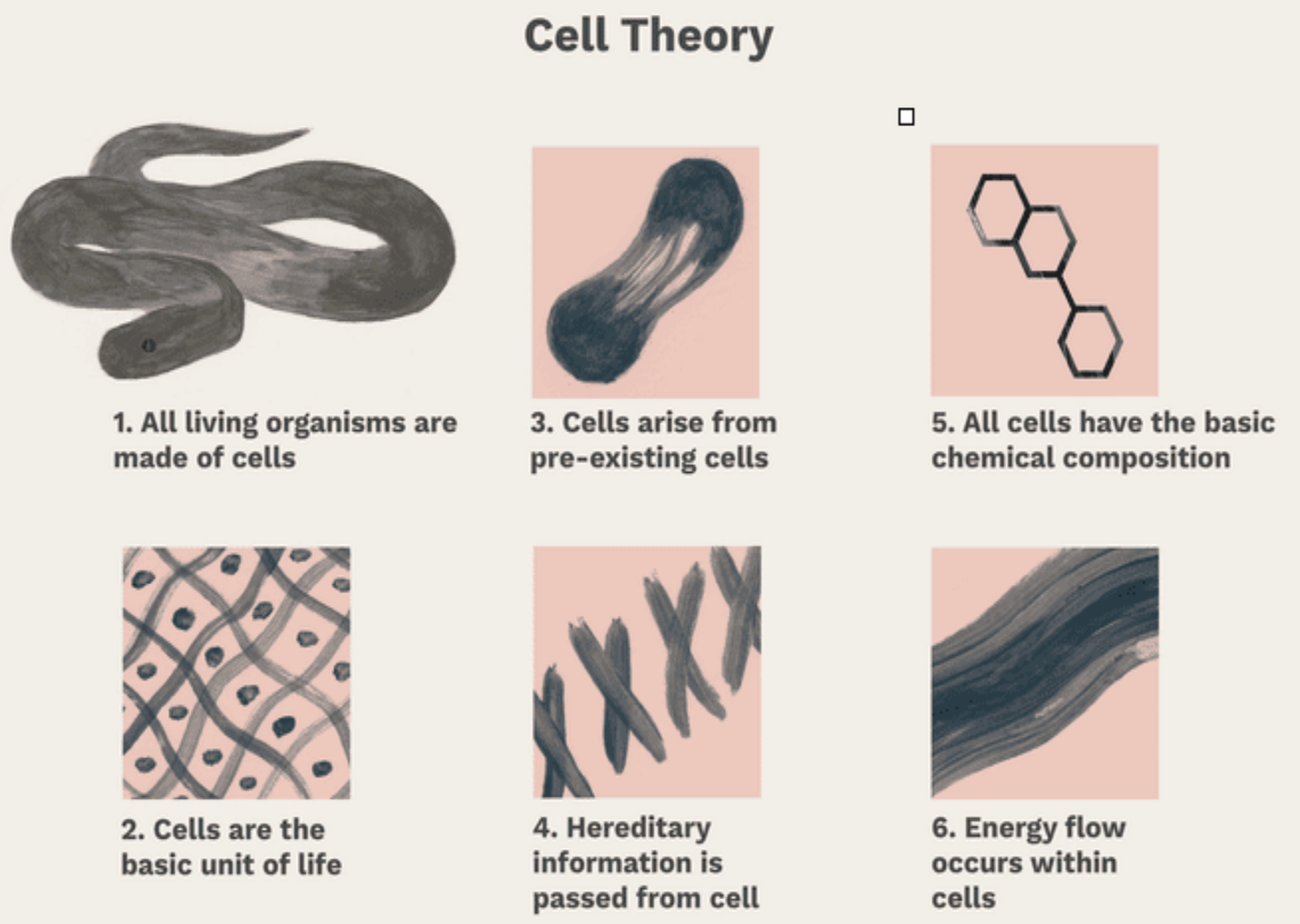
All living species on Earth are composed of cells.

A cell is the basic unit of life.

All cells arise from pre-existing cells.

A MODERN VERSION OF THE CELL THEORY WAS EVENTUALLY FORMULATED, AND IT CONTAINS THE FOLLOWING POSTULATES:

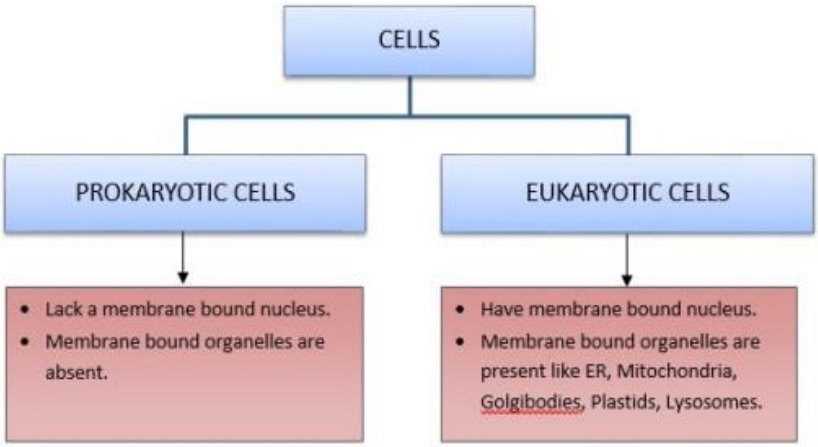
Energy flows within the cells.
Genetic information is passed on from one cell to the other.
The chemical composition of all the cells is the same.



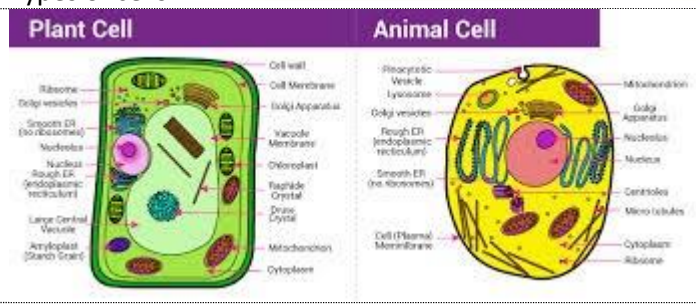
Functions of Cell:

A cell performs these major functions essential for the growth and development of an organism Important functions of cell are as follows:

1. Provides Support and Structure.
2. All the organisms are made up of cells. They form the structural basis of all the organisms.
3. Facilitate Growth Mitosis.
4. In the process of mitosis, the parent cell divides into the daughter cells. Thus, the cells multiply and facilitate the growth in an organism.
5. Allows Transport of Substances.
6. Various nutrients are imported by the cells to carry out various chemical processes going on inside the cells.
7. The waste produced by the chemical processes is eliminated from the cells by active and passive transport. Small molecules such as oxygen, carbon dioxide, and ethanol diffuse across the cell membrane along the concentration gradient. This is known **as** **passive transport**. The larger molecules diffuse across the cell membrane through **active transport** where the cells require a lot of energy to transport the substances.



Types of cells:



Types of Cells:

Cells are similar to factories with different laborers and departments that work towards a common objective. Various types of cells perform different functions.

Based on cellular structure, there are two types of cells:

PROKARYOTIC CELLS

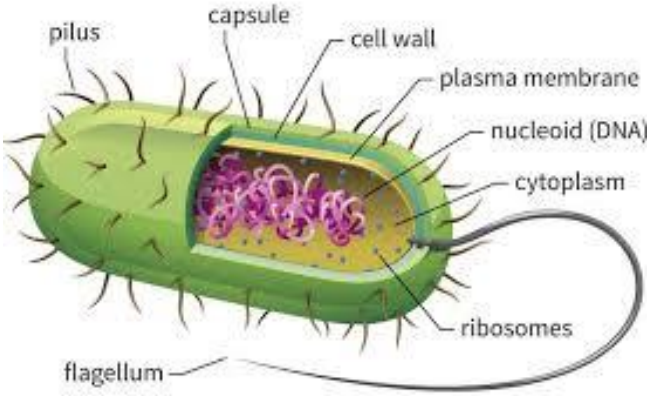
Prokaryotic cells have no nucleus. Instead, some prokaryotes such as bacteria have a region within the cell where the genetic material is freely suspended. This region is called the nucleoid.

They all are single-celled microorganisms. Examples include archaea bacteria and cyanobacteria.

The cell size ranges from 0.1 to 0.5 µm in diameter.

The hereditary material can either be DNA or RNA.

Prokaryotes reproduce by binary fission, a form of sexual reproduction.



EUKARYOTIC CELLS

Eukaryotic cells are characterised by a true nucleus.

The size of the cells ranges between 10–100 µm in diameter.

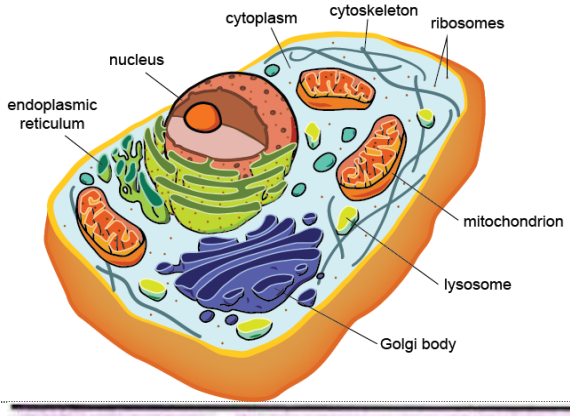
This broad category involves plants, fungi, protozoans, and animals.

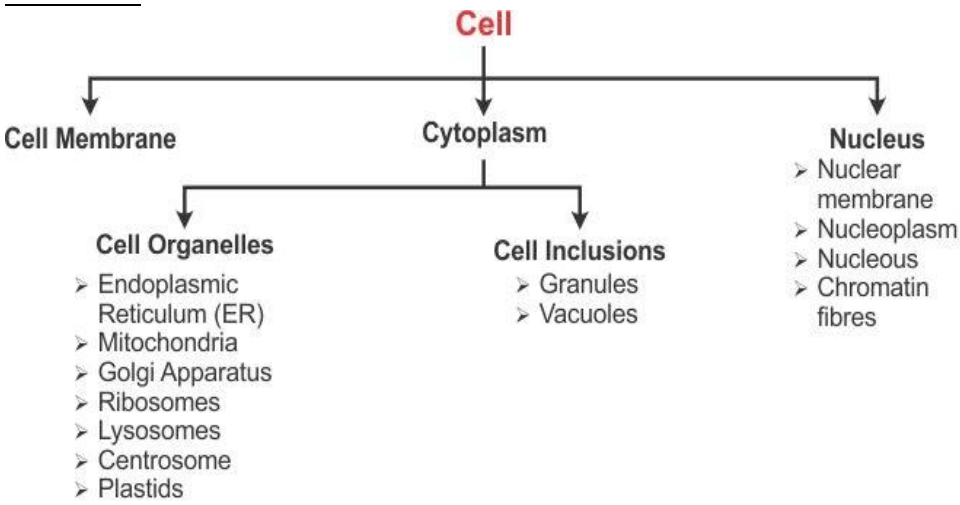
The plasma membrane is responsible for monitoring the transport of nutrients and electrolytes in and out of the cells. It is also responsible for cell to cell communication.

They reproduce sexually as well as asexually.

There are some contrasting features between plant and animal cells. For eg., the plant cell contains

chloroplast, central vacuoles, and other plastids, whereas the animal cells do not.



Comparison between prokaryotic and eukaryotic cells

Character	*Prokaryotic cell*	*Eukaryotic cell*
Size	Small (generally 1-10 µm)	Large (generally 10-100 µm)
Cell membrane	Cell is enveloped by a rigid cell wall	Cell is enveloped by a flexible plasma membrane
Sub-cellular organelles	Absent	Distinct organelles are found (e.g. mitochondria, nucleus, lysosomes)
Nucleus	Not well defined; DNA is found as nucleoid, histones are absent	Nucleus is well defined, surrounded by a membrane; DNA is associated with histones
Energy metabolism	Mitochondria absent, enzymes of energy metabolism bound to membrane	Enzymes of energy metabolism are located in mitochondria
Cell division	Usually fission and no mitosis	Mitosis
Cytoplasm	Organelles and cytoskeleton absent	Contains organelles and cytoskeleton (a network of tubules and filaments)